

CRITICAL METALS AND THE ENERGY TRANSITION

An analysis of Europe's strategic dependence

“To what extent does Europe's energy transition create a new strategic dependence on critical raw materials?”

Academic report

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Introduction

Over the past decade, the European Union has been driving a profound transformation of its energy system. The rollout of electric vehicles, storage batteries, wind and solar power, and modernised electricity grids is intended to reduce the continent's dependence on imported hydrocarbons. This transition is generally presented as a way to strengthen European energy autonomy.

Low-carbon technologies, however, rely on mineral raw materials whose production and processing are concentrated in a small number of countries. Unlike hydrocarbon deposits, which are relatively dispersed worldwide, several strategic metals and minerals — lithium, cobalt, rare earths, graphite — are extracted or refined in only a few geographic areas. The energy transition therefore does not necessarily remove Europe's external dependence: it shifts it, from fossil fuels to critical raw materials.

The issue at stake is therefore not only an energy one: it is also industrial and geopolitical. This is the dimension this report sets out to document.

This observation led the European Union to adopt, on 11 April 2024, the regulation on critical raw materials (the Critical Raw Materials Act), which sets targets for security of supply, diversification and industrial development by 2030 (EU Regulation 2024/1252). This framework is the starting point of the present report.

The research question addressed here is the following: *to what extent does Europe's energy transition create a new strategic dependence on critical raw materials?*

To answer this question, the study does not cover all seventeen strategic raw materials identified by the European Union. It focuses on eight materials directly mobilised by energy-transition technologies: battery-grade lithium and graphite, cobalt, battery-grade nickel, rare earths used in permanent magnets, copper, silicon metal and platinum-group metals (PGMs). This narrower scope allows for a more in-depth analysis than an exhaustive inventory of all seventeen materials would.

The approach combines data from four institutional sources (the EU regulation, the European Commission study, the USGS report, and the IEA report), a scoring grid built around five vulnerability criteria, and a strategic analysis of the geographic location of value chains. The report is organised into five parts: the European regulatory framework, the data and methodology used, the scoring results, a strategic analysis of the risks identified, and recommendations.

Part 1 — The EU framework for critical raw materials

Regulation (EU) 2024/1252 of 11 April 2024, known as the Critical Raw Materials Act (CRMA), establishes the legal framework through which the European Union intends to secure its supply of critical raw materials. The text starts from a simple observation: several materials that are indispensable to low-carbon technologies and to certain strategic sectors (digital, defence, aerospace) are imported almost entirely, often from a very small number of supplier countries.

The regulation pursues four main objectives. The first is to secure supply, in particular through monitoring the most sensitive supply chains. The second is to diversify suppliers, in order to reduce dependence on

a single country or a single geographic area. The third is to develop European extraction, processing and recycling capacity, through the designation of strategic projects benefiting from accelerated administrative procedures. The fourth is to encourage the recycling of critical raw materials already in circulation in the European economy, notably those contained in batteries, permanent magnets or electronic equipment at the end of its life.

To give concrete substance to these objectives, the regulation sets indicative benchmarks to be reached by 2030, summarised in the table below (EU Regulation 2024/1252).

2030 target	Indicative benchmark set by the regulation
Domestic extraction	At least 10% of the EU's annual consumption
Domestic processing	At least 40% of the EU's annual consumption
Recycling	At least 25% of the EU's annual consumption
Diversification	No more than 65% of the EU's annual consumption of each strategic material, at any given stage, from a single third country

Source: Regulation (EU) 2024/1252 of 11 April 2024.

The regulation distinguishes between two categories of materials. A critical raw material is one whose economic importance and supply risk exceed certain thresholds, based on a regular assessment by the European Commission. A strategic raw material is a narrower subset — seventeen materials in total — judged indispensable to energy-transition and digital technologies as well as to defence, and for which projected demand growth exceeds anticipated supply capacity. All strategic materials are therefore critical, but not all critical materials are strategic.

This report does not cover all seventeen of these strategic materials. It focuses on eight of them, selected because they are directly mobilised by energy-transition technologies: battery-grade lithium and graphite, used in lithium-ion battery cells; cobalt and battery-grade nickel, which are part of certain battery chemistries; rare earths used in the permanent magnets of wind turbines and electric motors; copper, essential to all electrical infrastructure; silicon metal, the basis of photovoltaic panels and certain semiconductors; and PGMs, used notably in hydrogen technologies and fuel cells. This narrower scope allows for a more detailed analysis than an exhaustive inventory would, without straying from the heart of the subject: the materials that determine the success of Europe's energy transition.

Part 2 — Data and methodology

Sources used

The analysis draws on four institutional sources, chosen for their complementarity.

Regulation (EU) 2024/1252 provides the legal framework and the official list of the European Union's critical and strategic raw materials. It serves as a reference for justifying the scope of the study and for situating each material within its regulatory status.

The European Commission's Study on the Critical Raw Materials for the EU 2023 is the main source of quantitative data on the geographic origin of raw materials (European Commission, 2023). It contains

two tables used in this report in particular: a table of the main global suppliers by material and by stage of the value chain (extraction or processing), and an equivalent table focused on European Union suppliers. These two tables make it possible to distinguish global geographic concentration from Europe's own dependence, which can differ significantly: a market that is not very concentrated at the global level can still be a source of strong European dependence if the Union sources almost exclusively from a single partner.

The USGS Mineral Commodity Summaries 2025 supplements this data with technical fact sheets by material, including global mine production by country for 2024. It indicates, for example, that global natural graphite production remains concentrated in China at around 78% in 2024, and that rare earth and PGM production remains strongly associated with a small number of producing countries (USGS, 2025).

The International Energy Agency's Global Critical Minerals Outlook 2025 provides additional insight into demand dynamics. It indicates that lithium demand grew by close to 30% in 2024, while demand for nickel, cobalt, graphite and rare earths grew by 6 to 8% over the same period, driven by electric vehicles, stationary storage, renewable energy and grid development (IEA, 2025). The IEA also notes that diversification of supply chains remains slow, and that refining and processing are generally more geographically concentrated than mining itself (IEA, 2025) — a central point for understanding European vulnerability beyond the question of geological reserves alone.

The five scoring criteria

Each material is assessed against five criteria, rated from 1 to 5: the material's importance for the energy transition, the geographic concentration of extraction, the geographic concentration of processing or refining, the degree of European dependence on external suppliers, and the difficulty of substituting the material with another material or technology.

These five criteria are weighted to produce a final score out of 100 points, using the following formula:

$$\text{Score} = (\text{Importance} \times 5) + (\text{Extraction concentration} \times 4) + (\text{Processing concentration} \times 4) + (\text{EU dependence} \times 4) + (\text{Substitution difficulty} \times 3)$$

Criterion	Weight	Maximum points
Importance for the energy transition	× 5	25
Extraction concentration	× 4	20
Processing concentration	× 4	20
EU dependence	× 4	20
Substitution difficulty	× 3	15

The resulting score is then interpreted using a four-level scale: critical (80 to 100 points), high (65 to 79 points), medium (50 to 64 points) and low (below 50 points).

Scope and limitations of the method

It is worth being clear about what this grid can and cannot do. The scoring grid does not claim to objectively measure the absolute vulnerability of each material. It is an analytical tool that allows risk profiles to be compared using consistent criteria — not a measuring instrument in the strict sense.

Several limitations follow from this methodological choice. First, the 1-to-5 rating necessarily simplifies complex realities: two materials rated 5 on the same criterion may in fact present slightly different degrees of risk that a discrete scale cannot capture precisely. Second, the available data is not always symmetrical across the three dimensions of extraction, processing and European dependence: some sources document extraction well but remain patchy on processing, or vice versa, which introduces a degree of uncertainty into the scoring. This is notably the case for copper, whose extraction is relatively well documented and diversified, while some refining circuits are harder to characterise precisely.

In addition, some of the categories used actually group together several distinct metals. Rare earths cover both light and heavy elements whose availability differs considerably; PGMs group together several metals (platinum, palladium, rhodium, iridium, ruthenium) whose dominant producing country is not always the same. Treating these categories as single blocks therefore smooths over real internal differences. Finally, the available data on European suppliers is incomplete in the sources used for certain materials, particularly cobalt and PGMs; this report flags this whenever it is the case rather than filling the gaps with unsourced estimates. The score should therefore be read as a synthetic indicator of vulnerability, not as an exact measurement.

Part 3 — Results

The ranking produced by the scoring grid clearly distinguishes two groups of materials: five materials classified as critical (score of 80 or above) and three materials classified as high risk (score between 65 and 79). None of the eight materials studied scores below 65, which confirms that the selection made in the introduction does indeed cover materials with significant risk.

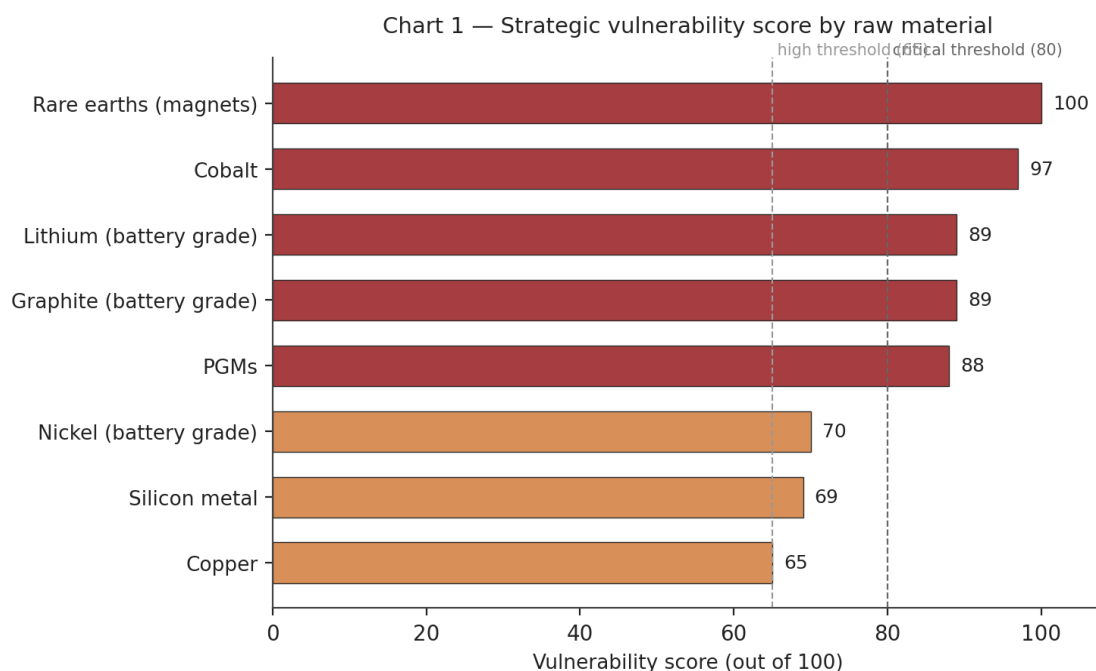


Chart 1 — Strategic vulnerability score by raw material

Rare earths used in permanent magnets obtain the maximum score of 100 out of 100: they accumulate the maximum number of points on each of the five criteria. This results from a combination of factors that reinforce one another rather than from a single determinant. Rare earth mining is concentrated, but it is above all their separation and processing into oxides and then magnets that rests almost entirely on Chinese capacity (European Commission, 2023; USGS, 2025). On top of this comes a particularly strong substitution difficulty: there is currently no alternative material capable of reproducing, at a comparable cost, the magnetic properties of the magnets used in wind turbines and electric motors.

Cobalt follows closely, with a score of 97, for rather different reasons. Available data indicates that its extraction is concentrated at more than 60% in the Democratic Republic of Congo, and that its refining is similarly concentrated (European Commission, 2023). The sources used, however, do not specify a leading European supplier for this material, which limits any reading of strictly European dependence beyond the rating assigned. Substitution is also considered difficult, even though technological developments — notably the development of cobalt-free battery chemistries — could eventually ease this dependence; the IEA report remains cautious about the pace of this shift (IEA, 2025).

Battery-grade lithium and graphite come next, tied with a score of 89. For lithium, the risk lies less in extraction, which is relatively diversified, than in processing into battery-grade compounds, a stage where China holds a dominant position according to European Commission data (European Commission, 2023). Battery-grade graphite shows an even more concentrated profile: China dominates both extraction and processing, with the USGS putting China's share of global natural graphite production at around 78% in 2024 (USGS, 2025; European Commission, 2023).

PGMs obtain a score of 88, which may be surprising given that their use in the energy transition remains more limited — mainly hydrogen technologies and fuel cells — than that of lithium or copper. This high score is explained by extreme production concentration, shared between South Africa for several PGMs and Russia for palladium (USGS, 2025), and by a genuine substitution difficulty in catalytic applications. It

should be noted, however, that the available European data remains incomplete for this material in the sources used, which calls for particular caution when reading the EU dependence rating. A more specific use does not, however, mean a lower risk: it simply means that a supply disruption would affect a smaller number of value chains, without this reducing the probability or the intensity of the risk itself.

By contrast, battery-grade nickel, silicon metal and copper are classified as high risk rather than critical, with scores of 70, 69 and 65 respectively. Nickel stands out for its more moderate European dependence than the other materials (a rating of 2 out of 5): the data used indicates that Russia is only one supplier among several in European sourcing, without dominating the chain the way China does for other materials (European Commission, 2023). Silicon metal shows relatively diversified extraction but concentrated processing, notably in China; its European dependence and substitution difficulty nonetheless remain moderate. Copper, finally, appears as the least vulnerable material in the group: its extraction is spread across several major producing countries, led by Chile (European Commission, 2023), and while its processing shows a more marked concentration, its European dependence remains limited in the available data. Copper illustrates a point worth highlighting here: a material can be absolutely indispensable to the energy transition — its importance is rated at the maximum, 5 out of 5 — without necessarily presenting the highest risk profile, provided the concentration and dependence criteria are more favourable.

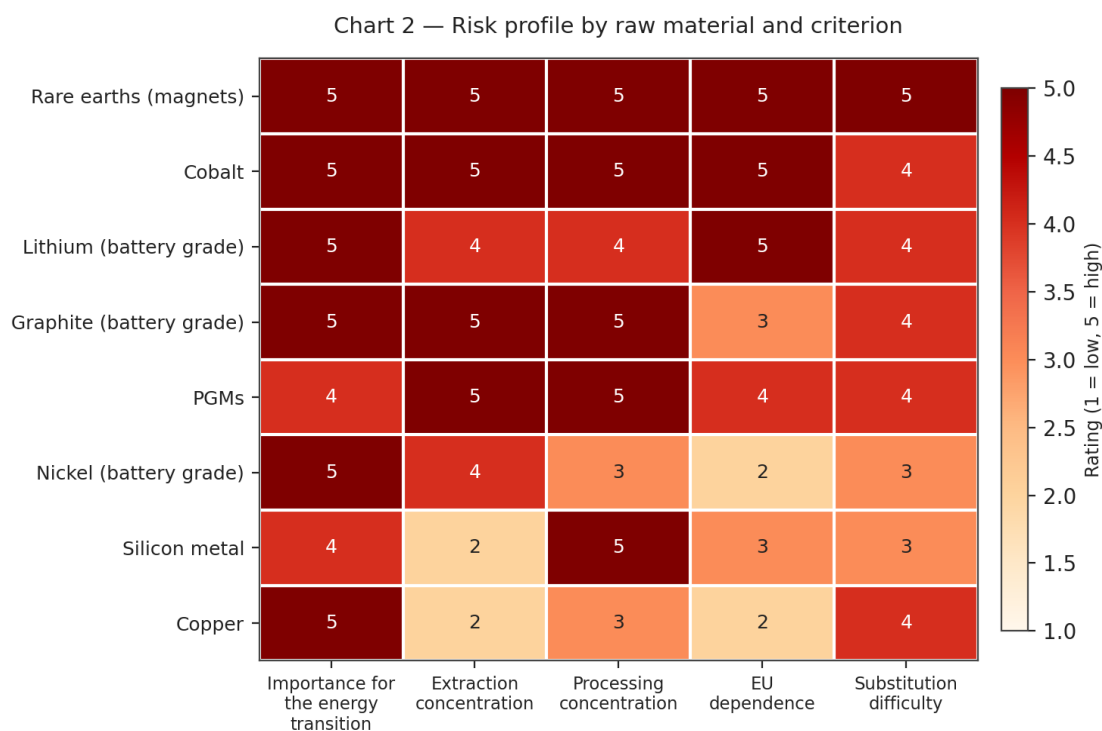


Chart 2 — Risk profile by raw material and criterion

Chart 2 breaks down, for each material, the rating obtained on each of the five criteria. It shows that the overall score conceals different risk profiles across materials. Copper and silicon metal stand out for a low rating on extraction concentration, which limits their overall score despite more problematic processing or dependence on certain points. Nickel shows the most contrasted profile in the group: its maximum rating on energy importance contrasts with markedly lower ratings on European dependence and processing concentration, reflecting a European supply situation that is relatively less tight than for the

other battery materials. Cobalt and rare earths, by contrast, show a near-uniform profile of high ratings across all criteria, reflecting a vulnerability that does not rest on a single link in the chain but on the chain as a whole.

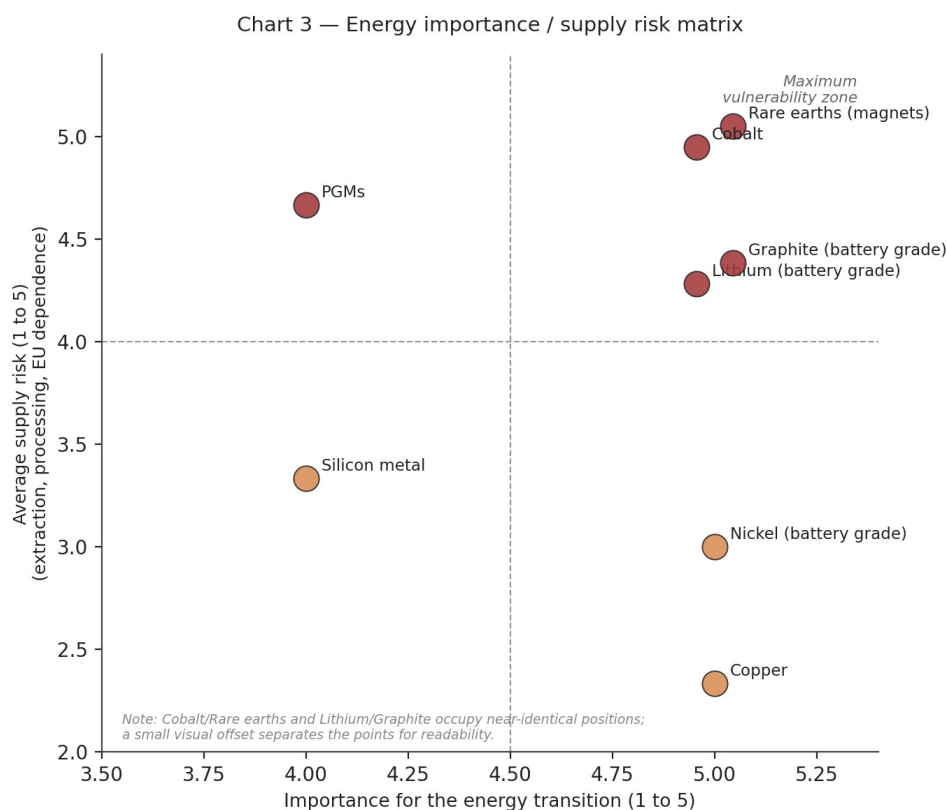


Chart 3 — Energy importance / supply risk matrix

Chart 3 plots each material's importance for the energy transition against an average supply risk, defined here as the average of the three criteria directly related to the supply chain: extraction concentration, processing concentration, and EU dependence. Substitution difficulty is deliberately left out of this calculation: it is already counted in the final score, but it describes technological room for manoeuvre rather than a feature of supply itself.

On this basis, the chart reveals a tight cluster of materials in the zone of high importance and high risk: rare earths and cobalt show the highest average supply risk (5 out of 5), closely followed by lithium and graphite (just above 4 out of 5). These are the materials for which a supply shock would combine a high probability with significant consequences for the energy transition. PGMs sit slightly apart on the importance axis, owing to a more limited use, but retain one of the highest average supply risks on the chart, close to 4.7 out of 5. Nickel and silicon metal occupy an intermediate position, with an average risk of around 3 out of 5. Copper, finally, occupies a distinctive position: its importance for the energy transition is maximal, but its average supply risk is the lowest in the group (just above 2 out of 5), which sets it clearly apart from the other materials studied and confirms the observation made at the start of this part.

Part 4 — Strategic analysis

The results above show that the risk associated with critical raw materials does not stem solely from the geological scarcity of resources. At the global level, reserves of lithium, nickel or copper are not exceptionally scarce; the risk stems above all from the geographic concentration of processing and refining capacity, which is often more pronounced than that of extraction itself. The IEA makes this point explicitly in its 2025 report (IEA, 2025): the intermediate stage of the value chain — processing ores into materials usable by industry — is often the most vulnerable link, even though it receives less public attention than mining.

China occupies a central position in several of these value chains (European Commission, 2023; USGS, 2025), not necessarily because it holds the largest geological reserves, but because it has built up, over several decades, industrial processing capacity in lithium refining, graphite anode production, rare earth separation, and industrial-grade silicon metal production. This position results mainly from long-term industrial choices and investment rather than from a simple natural advantage, which makes European dependence all the more structural: it probably cannot be resolved through diversification of extraction sources alone, but requires rebuilding processing capacity elsewhere, which takes time and substantial investment.

Other, more specific geographic risks are worth mentioning, even though the available data does not always allow them to be documented with the same precision. The Democratic Republic of Congo dominates global cobalt extraction (European Commission, 2023), in a context where governance and political stability concerns add to the risk of geographic concentration. South Africa and Russia jointly dominate PGM production (USGS, 2025), two countries whose political contexts and relations with the European Union differ considerably, which complicates a uniform supply policy for this group of metals — all the more so as the available European data on this point remains partial. Chile appears as the leading European supplier of lithium (European Commission, 2023), a situation that is a priori less politically sensitive than dependence on China, but which remains a dependence on a single country for a material judged critical.

These elements converge on a single conclusion: Europe is vulnerable not because it lacks resources on its own territory for all these materials, but because it depends on industrial chains largely located outside its borders, at stages — processing in particular — where it has little capacity of its own. Europe's energy transition is therefore not only a technological or energy transition: it is also a material transition, one that shifts part of Europe's external dependence from fossil fuels to critical raw materials, without necessarily reducing it to the same degree as is sometimes suggested in public debate.

Part 5 — Recommendations

The findings above point to differentiated priorities depending on the material and the stage of the value chain concerned.

1. Build European processing and refining capacity

Since the risk identified lies mainly at the processing stage rather than at extraction, European industrial efforts would be better directed primarily at lithium refining, rare earth separation and graphite anode production, rather than at mining alone.

2. Diversify suppliers

Partnerships with Canada, Australia, and certain countries in Africa and Latin America, alongside European producers, would help reduce dependence on a single supplier country, particularly for lithium, nickel and rare earths.

3. Strengthen recycling

Recycling of batteries, permanent magnets and PGMs provides a secondary source of supply that does not depend on mining geography. Its development would contribute directly to the regulatory target of 25% of consumption covered by recycling by 2030 (EU Regulation 2024/1252).

4. Support substitution where technically feasible

Certain technological developments, notably low-cobalt or cobalt-free battery chemistries, can reduce dependence over the long term. This lever nonetheless remains limited for the materials where substitution is judged most difficult, such as the rare earths used in permanent magnets.

5. Accelerate European strategic projects without neglecting social and environmental concerns

The designation of strategic projects provided for by the regulation must be accompanied by consistent environmental and social requirements, or it risks shifting risks rather than reducing them.

6. Establish ongoing monitoring of supply chains

Regular tracking of production, processing and European dependence data, along the lines of the tables used in this report, would allow the vulnerability assessment for each material to be updated periodically.

7. Differentiate policy by material

Copper, whose average supply risk is the lowest of the group studied, does not call for the same priorities as cobalt or rare earths, whose vulnerability rests on every link in the value chain. A uniform policy would risk misallocating public effort.

Conclusion

The question posed in the introduction was to what extent Europe's energy transition creates a new strategic dependence on critical raw materials. The scoring results and the strategic analysis that accompanies them allow for a nuanced answer.

Of the eight materials studied, five show a critical level of vulnerability: rare earths used in permanent magnets, cobalt, battery-grade lithium and graphite, and PGMs. Battery-grade nickel, silicon metal and copper show a high but not critical risk, which shows that a material's importance for the energy transition is not, on its own, sufficient to determine its level of vulnerability.

The strategic analysis carried out in Part 4 provides an essential additional insight: European dependence is less a matter of the geological quantity available worldwide than a matter of industrial control over

value chains, particularly at the processing and refining stage. China's dominant position in several of these chains results from long-term industrial choices, not from a simple advantage in natural resources.

This leads to a broader conclusion: the energy autonomy that the European Union is seeking through its low-carbon transition cannot be built without equal attention to material and industrial autonomy. Reducing dependence on fossil fuels while allowing an equivalent, or even more concentrated, dependence on critical raw materials to develop would amount to shifting a vulnerability rather than resolving it.

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Appendices

Appendix 1 — Full scoring table

Raw material	Importance	Extraction	Processing	EU dependence	Substitution	Score	Level
Lithium (battery grade)	5	4	4	5	4	89	Critical
Cobalt	5	5	5	5	4	97	Critical
Nickel (battery grade)	5	4	3	2	3	70	High
Graphite (battery grade)	5	5	5	3	4	89	Critical
Rare earths (magnets)	5	5	5	5	5	100	Critical
Copper	5	2	3	2	4	65	High
Silicon metal	4	2	5	3	3	69	High
PGMs	4	5	5	4	4	88	Critical

Notes: each criterion is rated from 1 (low risk) to 5 (high risk). Score out of 100 points using the formula presented in Part 2.

Appendix 2 — Source data (global and European suppliers)

Material	Leading global supplier	Stage	Leading EU supplier	Stage
Lithium	China — 56%	Processing	Chile — 79%	Processing
Cobalt	DRC — 63%	Extraction	Not available	Extraction
Nickel	China — 33%	Processing	Russia — 29%	Processing
Natural graphite	China — 67%	Extraction	China — 40%	Extraction
Copper	Chile — 28%	Extraction	Poland — 19%	Extraction
Silicon metal	China — 76%	Processing	Norway — 33%	Processing
Light rare earths	China — 85%	Processing	China — 85%	Processing
Heavy rare earths	China — 100%	Processing	China — 100%	Processing
PGMs	South Africa / Russia (by metal)	Extraction/processing	Incomplete data	Processing

Primary source: European Commission, Study on the Critical Raw Materials for the EU 2023, Tables A and B; supplementary source: USGS, Mineral Commodity Summaries 2025.

Appendix 3 — Score calculation formula

Score = (Importance for the energy transition × 5) + (Extraction concentration × 4) + (Processing concentration × 4) + (EU dependence × 4) + (Substitution difficulty × 3)

Vulnerability scale: critical (80 to 100 points), high (65 to 79 points), medium (50 to 64 points), low (below 50 points).